



Storing the sun's energy

Scientists may be able to mimic plants' photosynthesis in order to store the sun's energy. <http://digit.in/sep20-51>



Safer wind turbines for birds

A simple design change to wind turbines could make them much safer for avians - a paint job. <http://digit.in/sep20-52>



Credit: B Anasori, Drexel

MXenes

The next supermaterial

Aditya Madanapalle | aditya@digit.in

In 2010, researchers at Drexel University created a new class of nanomaterials, and published their findings a year later. The material was based on ductile and machinable ceramics known as MAX phases, and was called MXenes, pronounced as “maxene”. It was a 2D material, such as the better known graphene, and could be manipulated in a similar way, such as creating nanotubes by rolling the sheets up into tubes lesser than the thickness of a human hair. The first MXene was obtained by placing titanium-aluminum carbide in hydrofluoric acid, to selectively remove the aluminum, through a process called as exfoliation. They are

sheets of carbides and nitrides of transition metals, such as titanium and molybdenum. The resulting material had properties strikingly different from the 3D counterparts. The MXenes were good conductors of electricity, even though the ceramics were not. The creation of the first MXene actually opened up the possibility of creating a number of related 2D materials. These materials are strong, are very good at molecular filtration, and offer high electrical conductivity, all of which are beneficial properties. Since then over 25 different MXenes have been created, all with a number of potential applications. Simulations using computers have indicated that there may be millions of MXenes that are still to be discovered, and the researchers are in the process of identifying the ones that are stable, easy to manufacture, and can be used at scale.

One of the earliest potential applications to be discovered was that of energy storage devices in batteries.

Researchers showed that lumps of the MXene material could be made as soft and malleable as clay. This means that the material can be easily moulded into any shape or size, as is required by the application. Additionally, the clay has many properties similar to water, and can hold water, unlike graphene, which is hydrophobic. You can literally use a rolling pin to shape out electrodes or capacitors. The discovery was made when a student used fluoride salt and hydrochloric acid to remove the aluminum instead of the hydrofluoric acid, which is very toxic, and then used water to wash away residual chemicals and stop the reaction. Instead of finding the layers of MXene sheets, the resulting flakes were in the form of a clay. The researchers were looking for a simpler and less hazardous method to make MXene sheets, so they continued to work with the clay, and rolled it into flat films, which worked. The conductive properties of the material was maintained. They found that the capacity to hold charge of the MXene clay was actually three times more than the MXene sheets produced before, and as such were ideal candidates for making batteries, that could power anything from handheld electronics to vehicles. In tests conducted by the researchers, the MXene showed higher capacitance for volume than most other materials, and was able to hold charge after more than 10,000 charge and discharge cycles. As the clay was soluble in water, it could just be applied to other materials, where it would solidify.

This was followed up by a number of refinements in how MXenes can be possibly used. Research showed that a battery could be charged or discharged in a matter of seconds, which would allow the batteries of the future to be ready to go very rapidly. The researchers were also able to make a spray on solution, that could be used to make antennas for example, that could be applied to almost any material. The resulting layer of MXene is transparent, so for example, it could be used on the windows of autonomous